ALRS v2: Google ADK 4-Tier Swarm Architecture & Execution Topology

T1 **Tier 1: Orchestration & Reasoning** (gemini-2.5-flash, Temp=0.2)
T2 **Tier 2: Discovery & Execution** (gemini-3.5-flash-lite, Temp=0.7)
D **Self-Healing / Diagnostics** (Global Error Delegation Protocol)
L **Iterative Loop Container** (ADK LoopAgent, Max 3 Iterations)

⚡ Tri-Layer State Persistence & Shared Blackboard: tool_context.state['project_state'] ◄ ► Atomic JSON Disk ◄ ► MySQL Relational Audit Store ⚡

👑 ProjectAgent (Root Swarm Orchestrator)
Model: **gemini-2.5-flash** | Tier: **Tier 1** | Temp: 0.2
- **Cognitive Role:** Sequential 5-Stage Lifecycle Gating & Session Supervision
- **Assigned Tools:** create_new_project, select_project, get_project_state, transition_to_stage, get_stage_status, update_stage_substate, post_to_message_board, read_message_board
- **Hydration Hook:** _auto_hydrate_state_callback (recursive before_model_callback)

🛠️ DebuggerAgent (Autonomous Self-Repair)
Model: **gemini-2.5-flash** | Tier 1 | Temp: 0.1
- **Protocol:** Global Error Delegation Protocol
- **Capabilities:** Sandboxed AST Syntax Verification, Traceback Triage, Patch Proposal & Test Suite Execution
- **Tools:** read_agent_code, fetch_error_logs, run_agent_tests, validate_and_propose_patch

RESEARCH SUB-SWARM (Stage 1: Research & Stage 2: Analysis)

🔬 ResearchAgent (Domain Coordinator)
Model: **gemini-3.5-flash-lite** | Tier 2 | Temp: 0.7
- **Role:** Manages Research Questions & Paper Library Growth
- **Tools:** add_research_question, list_research_questions, get_project_papers, get_top_authors, get_top_venues

🤖 ALRSAgent (Paper Discovery Specialist)
Tier 2 | gemini-3.5-flash-lite
- Bounded 1-Hop GraphSelect BFS Citation Traversal
- OpenAlex API Polite Pool Search (cites / cited_by)
- Candidate Appraisal into 3 Archetypes (Found/Contemp/Bridge)
- 768-dim Gemini MRL Vector Embeddings

🔍 GapFinderAgent (Analysis & Centrality Specialist)
Tier 2 | gemini-3.5-flash-lite
- Pairwise Cosine Similarity Matrix Evaluation
- Uncovered RQ Deficit Detection & Gap Formulation
- 3-Tier Centrality Engine (Neo4j GDS / NetworkX / In-Degree)
- Hybrid Relevance Ranking (Score = 0.70 Sim + 0.30 PR)

📊 EvidenceSynthesizerAgent (Literature Synthesis)
Tier 2 | gemini-3.5-flash-lite
- Structured Findings & Benchmark Metrics Extraction
- Multi-Dimensional Cross-Study Comparison Matrix
- Qualitative Synthesis & Trade-off Analysis
- Epistemic Grounding of Claims & Limitations

PLANNING & WRITING SUBSYSTEM (Stage 3, Stage 4 & Stage 5)

📐 PlannerAgent (Outline Blueprints)
Model: **gemini-2.5-flash** | Tier 1 | Temp: 0.2
- **Role:** Formulates Narrative Arc & Section Blueprints
- **Requirements:** Assigns ≥3 Valid DOIs per Section
- **Styles:** Thematic, Chronological, Methodological
- **Tools:** get_project_summary, get_ranked_papers, generate_outline

🔄 WriterReviewerLoop (ADK LoopAgent, Max 3 Iterations)

📝 DraftingAgent (Scientific Prose Specialist)
Tier 2 | gemini-3.5-flash-lite
- High-throughput academic section drafting
- Rigorous inline (Author, Year) citation placement
- Grounded paper abstracts & findings injection
- Tools: get_paper_abstracts, draft_section

🛡️ ReviewerAgent (Quality & Grounding Gatekeeper)
Tier 1 | gemini-2.5-flash (Temp 0.0)
- **5-Dimension Rubric:** Coverage (0.25), Coherence (0.20), Synthesis (0.25), Tone (0.15), Completeness (0.15)
- **Strict Gate:** All 5 scores ≥ 4.0/5.0 & Zero Missing DOIs
- **GCDL Grounding Invariant:** HR = 0.0% (Zero Hallucinations)
- **Termination:** exit_loop (tool_context.actions.escalate=True)

✍️ WriterAgent (Writing Coordinator & Exporter)
Model: **gemini-3.5-flash-lite** | Tier 2 | Temp: 0.7
- **Role:** Supervises Draft-Review Loop & Publication Exports
- **Sub-Agent:** Coordinates WriterReviewerLoop
- **Export Tools:** export_latex, export_bibtex, export_citation_graph